

■ Vehicle Dashboard - Auto Start with systemd (Ubuntu Setup Guide)

This document provides a step-by-step guide to configure your **Vehicle Dashboard project** to automatically start all Docker containers (MySQL, Flask Backend, and React Frontend) using **systemd** on Ubuntu.

■ 1. Prerequisites

- 1 Ensure Docker and Docker Compose v2+ are installed on Ubuntu.
- 2 Ensure all Docker images are available locally or in your Docker Hub registry.
- 3 Ensure your `docker-compose.yml` file works manually before automating.

■ 2. Stop Existing Containers

Before enabling the system service, stop all currently running containers:

```
docker ps
docker stop react-dashboard backend-subscriber mysql-db
```

■■ 3. Create the systemd Service File

Create a new systemd service unit file:

```
sudo nano /etc/systemd/system/vehicle-dashboard.service
```

Paste the following configuration:

```
[Unit] Description=Vehicle Dashboard Docker Compose Stack Requires=docker.service
After=docker.service [Service] Type=oneshot
WorkingDirectory=/home/angad/DevOps/my_first_vehicle_dashboard
ExecStart=/usr/bin/docker compose -f docker-compose.prod.yml up -d
ExecStop=/usr/bin/docker compose -f docker-compose.prod.yml down RemainAfterExit=yes
[Install] WantedBy=multi-user.target
```

■ 4. Enable and Start the Service

Once the file is saved, reload systemd and start the service:

```
sudo systemctl daemon-reload
sudo systemctl enable vehicle-dashboard.service
sudo systemctl start vehicle-dashboard.service
```

■ 5. Verify Containers and Service

Check the service and running containers:

```
sudo systemctl status vehicle-dashboard.service
docker ps
```

■ 6. Common Management Commands

- Check service status: `sudo systemctl status vehicle-dashboard.service`
- Start service manually: `sudo systemctl start vehicle-dashboard.service`
- Stop all containers via service: `sudo systemctl stop vehicle-dashboard.service`

- Restart service: `sudo systemctl restart vehicle-dashboard.service`
- View logs: `sudo journalctl -u vehicle-dashboard.service -f`

■ 7. Optional Improvement

Add this line in each service of your docker-compose.yml to ensure auto-restart:

```
restart: always
```

■ You have successfully configured automatic container startup for your Vehicle Dashboard project using systemd!